
3AL Hand-in 3 2024

Due: Friday 19 April 11:00 pm

Please make sure all of your answers are clearly labelled.

1. Determine whether each statement is true or false. No explanation needed.

(a) If G is a finite group, then $|\text{class } a|$ divides $|G|$ for all $a \in G$.

(b) If $H \triangleleft G$ with $H \neq G$, then $H = N_G(H)$.

(c) If G is a group, then $\langle g \rangle \subseteq C_G(g)$ for all $g \in G$.

(d) If G is a group and $a, b \in G$, then ab and ba are conjugate.

[4 marks]

2. Let G be a group and $g \in G$. Suppose $|g| = n$. Prove that if g is the only element of G with order n , then $g \in Z(G)$.

[4 marks]

3. Let G be a group and T a set of representatives of the distinct conjugacy classes of G . Prove that $G = \bigcup_{g \in G} gTg^{-1}$.

[4 marks]

Maximum mark = 10

Available marks = 12